PSI Center for Scientific Computing, Theory and Data

LESTO TASK 3.4 | REPORT

The LESTO case ladder, explained

From the T-Flows benchmark to conservative wall exchange: model equations, verification and experimental comparison

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Cases 001–011: Bojan Ničeno • Cases 012–024: Jan Kren

Purpose of this report This report is prepared for Bojan Ničeno, developer of T-Flows and supervisor of the master's work by Lobresco that led to the benchmark paper [1]. It connects that shared starting point to the OpenFOAM case ladder: Bojan's carrier and transport cases 001–011, then Jan's wall-exchange and chemistry cases 012–024. The central questions are which equations have changed, how conservation is enforced, and what the comparisons actually establish. Case 022 checks reproduction of the T-Flows temperature model; case 023 tests the extended physical model against five experiments. These are different tests, not a ranking of CFD codes. The reported onset RMS error is 18 K versus about 35 K for the published models, conditional on the reconstructed experimental inputs.

1 The question

Lead–bismuth eutectic (LBE) is the liquid-metal coolant of some fast reactor designs. Fission products dissolved in it, iodine among them, can evaporate, and iodine does not leave as pure iodine: it combines with lead and bismuth into gases such as PbI_2 and BiI_3. Which compound forms matters, because each one condenses at a different temperature, and therefore ends up in a different place.

To study this, radiochemists use **thermosublimatography**: a long, narrow tube (a few millimetres wide, about a metre long) whose wall temperature falls steadily from about 1000 K at one end to near room temperature at the other. A small boat of material is heated near the hot end; a slow stream of helium carries the vapour downstream; each compound sticks to the wall where the wall is cold enough for it. Afterwards the tube is scanned, and the position of each deposit tells you its **deposition temperature**, T_{dep}.

The coupled problem is: the gas *flows* along the tube, the vapour *diffuses* sideways to the wall, and at the wall it *condenses* or *re-evaporates* depending on the local wall temperature and on what is already deposited there. The goal of the LESTO work in Task 3.4 is a computational fluid dynamics (CFD) model, built on OpenFOAM, that does all three and can be checked against the experiments.

Reading guide. The next section defines the equations and sign conventions. The ladder then identifies the case that checks each added capability. The final sections separate numerical verification, reproduction of the T-Flows benchmark and comparison with experiments. Here "frozen carrier" means that the helium fields are computed first and held fixed while the dilute reactive species evolve.

2 Equations behind the added physics

Transport on the frozen helium field

The steady carrier provides ρ, **u**, T and p. Each trace gas is advanced with

$$\underbrace{\frac{\partial(\rho Y_i)}{\partial t}}_{\text{storage}} + \underbrace{\nabla\cdot(\rho\mathbf{u}Y_i)}_{\text{advection}} = \underbrace{\nabla\cdot(\rho D_i\nabla Y_i)}_{\text{diffusion}} + S_i^{\text{rel}} + S_i^{\text{wall}}. \tag{1}$$

Y_i is species mass per unit helium-carrier mass; D_i is the molecular diffusivity. The two sources represent release from the boat and exchange with the wall, in $\mathrm{kg\,m^{-3}\,s^{-1}}$. The reactive Y_i do not have to sum to one. Their evolution does not feed back into carrier momentum or temperature. This is the trace-species approximation, and it is least secure close to a strongly evaporating boat.

From a prescribed onset to reversible exchange

The temperature model activates deposition using an imposed threshold. The HKS model instead compares the local vapour pressure with the equilibrium pressure of the condensed phase:

$$j_i = G_i[p_{\text{eq},i}(T_w) - p_{i,w}], \qquad G_i = C_e s_i \frac{2\alpha_i}{2-\alpha_i}\sqrt{\frac{M_i}{2\pi R T_w}}. \tag{2}$$

Here j_i is mass flux in $\mathrm{kg\,m^{-2}\,s^{-1}}$, positive *from the deposit into the gas*; G_i is in $\mathrm{s/m}$. M_i is molar mass in $\mathrm{kg/mol}$, R the gas constant, T_w interface temperature in K, and both pressures are in Pa. α_i is accommodation; C_e and s_i are dimensionless multipliers. The physical baseline sets all three to one.

If $p_{i,w} > p_{\text{eq},i}$, then $j_i < 0$ and vapour condenses. If $p_{i,w} < p_{\text{eq},i}$, then $j_i > 0$ and existing material evaporates. A bare wall cannot evaporate: release is capped by the available inventory. The equilibrium-pressure table selects the stable solid or liquid branch. Thus deposition temperature is an *output*, rather than an imposed threshold. The pressure law alone does not determine the profile: the gas must also reach the surface by diffusion.

Connecting the cell centre to the wall

The finite-volume unknown is $Y_{i,c}$ at the cell centre, not the wall pressure in Eq. (2). Define $\chi_{i,c} = \rho_c R T_c / M_i$, so that $p_{i,c} = \chi_{i,c} Y_{i,c}$. With wall distance d_f and face diffusivity $(\rho D_i)_f$, eliminating the unresolved wall concentration gives

$$\frac{1}{G_{i,f}^{\text{eff}}} = \frac{1}{G_i} + \frac{\chi_{i,c} d_f}{(\rho D_i)_f}, \qquad j_{i,f} = G_{i,f}^{\text{eff}}[p_{\text{eq},i}(T_w) - \chi_{i,c} Y_{i,c}]. \tag{3}$$

The two terms are kinetic and half-cell diffusion resistances in series. Even infinitely fast surface kinetics leave a finite diffusion limit. This is a connection to the resolved gas field, not an extra fitted sticking parameter. Case 016 checks precisely this change.

In that synthetic case the diffusion conductance is $D/d_f = 1\,\mathrm{m/s}$, against a kinetic velocity $G_i\chi_{i,c}/\rho_c = 63.7$–$107\,\mathrm{m/s}$. The combined value is only 0.985–0.991 m/s. Yet the 1 % onset moves by just 0.005 K, while the peak drops by 7.4 % and moves one bin downstream. An unchanged onset therefore does not imply an unchanged deposition profile.

Conservation during wall exchange

For a single gas–condensate pair, the same realised face flux updates both inventories:

$$S_{i,c}^{\text{wall}} = \frac{1}{V_c} \sum_{f \in c} A_f j_{i,f}, \qquad \Delta m_{i,f}^{\text{wall}} = -\Delta t \, A_f j_{i,f}. \tag{4}$$

A_f is wall-face area and V_c cell volume. The flux used here is the one actually realised by the gas solve after inventory bounds have been applied. Independently updating the wall with a predicted flux would not guarantee cancellation.

Once reactions change molecular identities, conservation must be checked for each element $k \in \{\text{Pb}, \text{Bi}, \text{I}\}$:

$$N_k^0 + N_k^{\text{rel}} + N_k^{\text{in}} - N_k^{\text{gas}} - N_k^{\text{wall}} - N_k^{\text{out}} = 0. \tag{5}$$

All N_k are amounts of the element in mol; inlet and outlet terms include the accounted boundary transport. The ledger reports the residual relative to the case's reference inventory. Round-off closure verifies bookkeeping of the represented processes. It does not establish the accuracy of transport, thermodynamic data or experimental assumptions.

Gas equilibrium preserves atoms, not molecules

For gas species i with ν_{ki} atoms of element k, local element concentrations are

$$b_k = \sum_i \nu_{ki} \frac{\rho Y_i}{M_i}. \tag{6}$$

The equilibrium step redistributes those atoms between the nine gases Pb, PbI, PbI_2, Bi, Bi_2, BiI, BiI_3, I and I_2:

$$\frac{p_i}{p^\circ} = K_{f,i}(T) \exp\left(\sum_k \nu_{ki} \lambda_k \right), \qquad \sum_i \nu_{ki} \frac{p_i}{RT} = b_k, \quad p^\circ = 10^5 \,\text{Pa}. \tag{7}$$

$K_{f,i}$ comes from the thermodynamic tables and the dimensionless element potentials λ_k are solved to satisfy the atom balances. The conversion back is $Y_i = M_i p_i / (\rho R T)$. This assumes instantaneous gas equilibrium, not measured reaction kinetics. In the implemented model the redistribution is requested at $T \geq 350\,\text{K}$; colder cells retain the transported composition. Homogeneous nucleation and aerosols are excluded.

Case 021 also adds shared metal deposits and the forward wall channel

$$3\text{BiI}(g) \longrightarrow 2\text{Bi}(\text{condensed}) + \text{BiI}_3(g). \tag{8}$$

Both sides contain three Bi and three I atoms. Different channels draw on the same physical Bi inventory. The forward-only channel is a modelling choice; it should not be read as a complete reversible surface mechanism.

> **What is inherited and what is added** The transport foundation is cases 001–011. The later ladder adds reversible, inventory-limited exchange, its diffusive connection to the wall, and element-conserving chemistry. The equations above follow the current manuscript and solver implementation; they make the changes explicit without requiring the reader to trace the source code.

3 The case ladder

A model like this has many ingredients: the gas flow, heat, variable gas properties, several chemical species, wall chemistry, gas-phase chemistry, and the real geometry and temperatures of each experiment. If you add them all at once and the answer is wrong, you do not know which ingredient is to blame. So we did the opposite. The folder `run/` of the LESTO repository holds 24 *cases*, numbered 001 to 024, and every case obeys the same four rules:

1. Its `readme.md` starts with *"This case is based on case N"*.
2. It changes **one thing**, and shows the changed lines.
3. It lists the **numbers to check** (e.g. "mass closure below 10^{-15}").
4. It says how to run it; cases 012–024 run with `./Allrun` and print a single word: `PASS`, `FAIL` or `NOTRUN`.

Figure 1 shows the result. The ladder climbs in four stages, which the next sections walk through: the helium flow (001–007), moving vapour through it (008–011), letting the vapour interact with the wall (012–021), and finally the comparisons with experiments and with another code (022–024). Anyone can enter at any rung, rerun it, and see the same numbers.

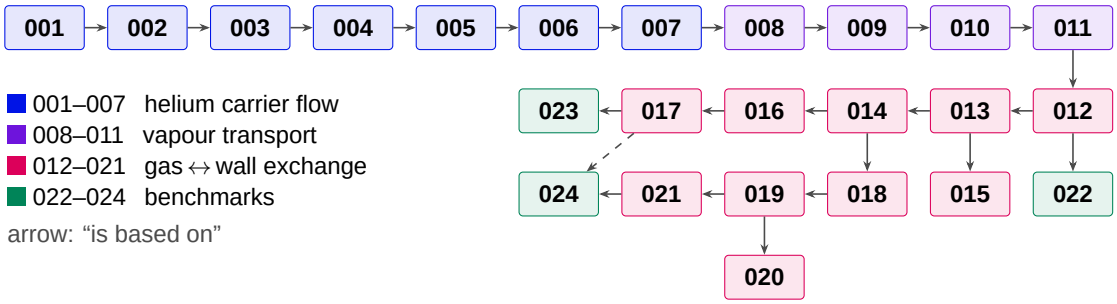


Figure 1: The 24 rungs. The trunk runs left to right along the top, then turns back along the second row. Side branches (015, 018–021, 022) add options that the main line does not need; 024 takes its chemistry from 021 and its realistic helium flow from 017 (dashed).

4 Climbing the ladder

Rungs 001–007: get the helium flowing

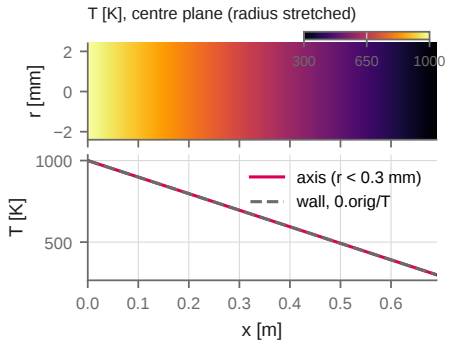
Bojan Ničeno

The first case is the plainest possible flow: helium through a pipe of 2.4 mm radius and 0.69 m length, at room temperature, on a mesh of 228,960 cells. Each next case adds one realistic touch: a parabolic velocity profile at the inlet (002) and inside the pipe from the start (003, which cuts the solver iterations from 388 to 222), the energy equation (004), a wall temperature that falls along the tube (005), a density that follows the ideal-gas law (006), and finally temperature-dependent viscosity and conductivity with a wall falling from 1000 K to 300 K (007, Fig. 2a). The saved result of 007 is the helium "carrier" that the next rungs read.

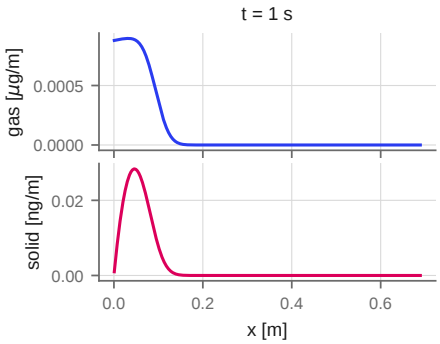
Rungs 008–011: move vapour through it

Bojan Ničeno

Now the flow is frozen and a new solver, `rhoFixedFlowFoam`, moves trace species through it. Case 008 sends PbI_2 vapour and an inert tracer with identical settings and checks that they stay identical (they do, to the last digit). 009 gives PbI_2 its real, temperature-dependent diffusivity, up to 25 times the tracer's on the hot side. 010 introduces a *solid* species, the deposit, and the interface through which chemistry will later talk to the flow. 011 couples them: what leaves the gas appears as solid on the wall, and the two sources add up to exactly zero (Fig. 2b).



(a) 007: the helium carrier. Top: temperature in the tube (radius stretched); bottom: the gas on the axis follows the wall from 1000 to 300 K.



(b) 011: after 1 s, PbI_2 gas enters from the left (top) and turns into deposit on the wall (bottom), mass for mass.

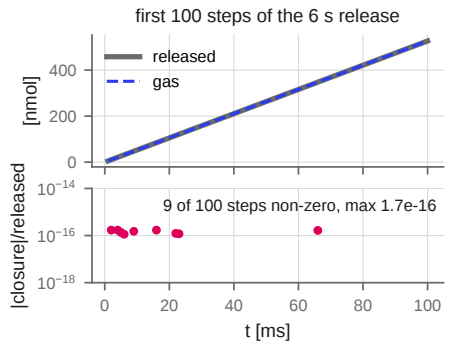
Figure 2: The first two stages: a correct carrier, then vapour that moves and deposits. (a) is drawn from the saved carrier of 007; (b) is a rerun of 011 to 1 s, made on 15 October 2026.

Rungs 012–021: wall exchange and chemistry

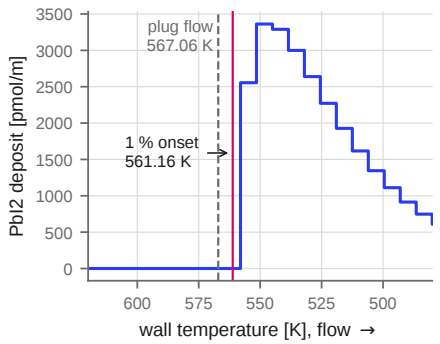
Jan Kren

Up to 011 the deposition rule was a placeholder. From 012 on, the cases add real wall physics, again one piece at a time:

- **012** The simple *temperature model* of Lobresco et al. [1]: vapour sticks once the wall is colder than a set T_{dep}. It also introduces a vapour source (a "boat") and the **mass ledger**: every step, every kilogram is accounted for. The unexplained remainder stays at about 10^{-16} of the released amount, i.e. at the limit of computer arithmetic (Fig. 3a).
- **013** The physical wall law (HKS): condensation and re-evaporation driven by the equilibrium vapour pressure of solid and liquid PbI_2, and a boat that evaporates its real inventory.
- **014** A cheap synthetic test channel (1000 K to 350 K) that runs in 10 s. Here the model's T_{dep} can be checked against the plug-flow hand calculation: 561.2 K against 567.1 K, within the 10 K gate (Fig. 3b).
- **016** The vapour has to *diffuse* to the wall before it can condense; this extra resistance turns out to control the uptake. It is used in every realistic run after this one.
- **017** The real helium flow and measured wall temperature of one Lobresco experiment, instead of the synthetic channel.
- **018–021** More chemistry: a second iodide (BiI_3), nine Pb–Bi–I gases that re-balance with each other everywhere in the tube (019; 020 does the same with GEMS3K), and metal vapour from LBE with shared deposits and wall reactions (021). In 021, all five deposit onsets land within 3.9 K of the hand calculation.



(a) 012: released and gas inventory (top) match; the unexplained remainder (bottom) is $\sim 10^{-16}$, machine round-off.



(b) 014: PbI_2 deposit along the synthetic channel. The model's onset (1 % rule, 561.2 K) sits 5.9 K from plug flow.

Figure 3: Two kinds of check used throughout the ladder: nothing is lost (a), and the answer agrees with a hand calculation where one exists (b).

5 The top rungs: does it match reality?

The last three cases do not add physics; they *test* it.

022 — reproduction of the T-Flows benchmark. This case returns to the temperature-model calculations in Lobresco's master's work with Bojan, published in Ref. [1], using Bojan's T-Flows code. In "paper mode" our model repeats their calculation: the gas level agrees with their Fig. 9 to 0.5 %, and all ten lines of their Table 2 agree within the resolution of the wall temperature table. The one visible difference, a deposit starting 2.6 cm earlier, disappears (to 0.9 mm) with the T-Flows parameter set; confirming which set generated the published figures is a specific point to resolve with Bojan using the original T-Flows inputs. The remaining 0.9 mm is an onset-position comparison, not proof that the entire profiles coincide.

023 — against five experiments. The five PbI_2 experiments of Lobresco et al. [1] differ in helium flow (5, 106 and 540 mL/min) and duration (6 to 300 s). We ran all five with the HKS wall law, the measured wall temperatures, and *no parameter adjusted to the data*, then applied the paper's own rule for T_{dep} (the point where the deposit reaches 1 % of its peak). Table 1 and Fig. 4a show the outcome: an RMS error of 18 K, against about 35 K for each of the two published model configurations. This is a comparison of complete modelling setups, not an isolated comparison of the OpenFOAM and T-Flows discretisations. A coarser mesh changes the answers by at most 0.3 K, and the mass ledger closes to 3.3×10^{-16}.

For a deposit profile $q(x)$, the reported onset is the first crossing of $0.01 \max q$ in the selected deposit region:

$$T_{\mathrm{dep},r} = T_w(x_{1\%,r}), \qquad E_{\mathrm{RMS}} = \sqrt{\frac{1}{5}\sum_{r=1}^{5}(T_{\mathrm{dep},r}^{\mathrm{model}} - T_{\mathrm{dep},r}^{\mathrm{exp}})^2}. \tag{9}$$

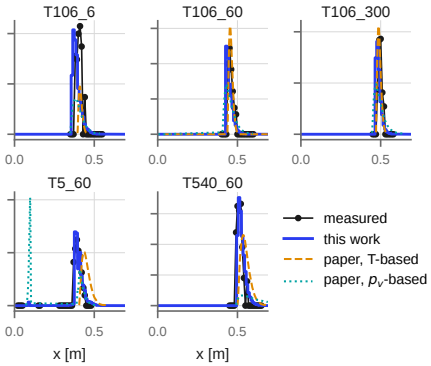
It is an onset statistic, not a peak-position or full-profile error. The low-flow case uses the main downstream deposit, as noted below.

Table 1: Case 023: predicted minus measured deposition temperature, production mesh. Experiment names: helium flow (mL/min) _ duration (s).

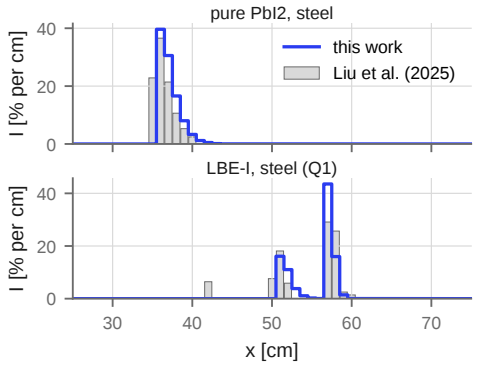
	T106_6	T106_60	T106_300	T5_60[a]	T540_60	**RMS**
This work (023)	+26 K	+19 K	+16 K	−11 K	−14 K	**18 K**
Lobresco et al., both models						35 K

[a] At 5 mL/min some vapour diffuses upstream and leaves a small deposit near the cold inlet; the value is for the main deposit downstream of the boat.

024 — the full Pb–Bi–I chemistry. Liu et al. [2] evaporated pure iodides and iodine-containing LBE into helium in steel and silica columns. Case 024 runs their configurations with the full chemistry of 021 on the realistic flows of 017, 60 s each on a mesh of 41,472 cells. For pure PbI_2 on steel the computed peak sits at 36 cm, 366.5 °C, against the measured 368 ± 18 °C, and for iodine from LBE the model reproduces the two separate deposit bands (Fig. 4b). Refining the mesh to 93,312 cells moves the profiles by 1.98 %.



(a) 023: deposit profiles of the five Lobresco experiments (arbitrary units) against tube position.



(b) 024: iodine deposit per cm, pure PbI_2 (top) and iodine from LBE (bottom), against Liu et al. (2025).

Figure 4: The benchmarks. Black dots and grey bars: measurements. Blue: this work. Dashed and dotted: the two models of the original paper.

6 What this gives us, and what it does not

What we have

- **A model with traceable verification checks.** Every capability of the solver has a rung that isolates it and a number that checks it. When something breaks later, `./Allrun-ladder 012-024` reruns all checks and points at the first rung that fails.
- **Nothing goes missing.** The recorded pair and element ledgers close near round-off in the cited cases. These checks address conservation, not the adequacy of the physical model.
- **A useful prediction.** Against five experiments the untuned model halves the error in T_{dep} of the published models (18 K vs 35 K RMS), and it reproduces the multi-band deposits of the LBE experiments.
- **A paper trail.** Cases 023 and 024 feed directly into the methods manuscript (Sec. 6.3, Table V, Fig. 5) and the ladder is the availability section of LESTO deliverable D38.

What we do not have yet

- Some inputs of the experiments are not published and are *assumed*: how fast the boat released its material, the reference conditions of the flow, and several positions. The 024 results are therefore labelled provisional.
- The model's deposit peaks sit 1 to 4 cm upstream of the measured ones in 023; the boat's obstruction of the flow and the dense vapour right next to it are not represented.
- The BiI_3 cases (018, 019, 021) need restricted thermodynamic tables that cannot be redistributed, and 015 and 020 need an optional GEMS3K build; without them those cases print `NOTRUN`.
- The mesh study of 024 shows small changes (1.98 %) but no formal grid-convergence index: the apparent orders of convergence are not consistent.

Points to discuss with Bojan

1. Confirm the original T-Flows inputs for Figs. 8–10: the imposed onset temperature, wall-layer thickness and rate parameters. This would resolve the case-022 onset discrepancy at its source.
2. Review the HKS units and pressure convention in both implementations, and compare the same interface law before attributing differences to the flow solver. The physical runs here use SI units and Eq. (3).
3. Prioritise the remaining input uncertainties: boat release history, flow reference conditions and boat position. Compare complete deposit profiles as well as the 1 % onset when assessing improvements.

> **How to try it** Every case lives in `run/<NNN-name>/`. Read its `readme.md`, then run `./Allrun` (cases 012–024; outputs go into a git-ignored `work/` folder) and look for PASS. The cheap cases take seconds to minutes; 017 about 8 minutes; the full 023 and 024 studies take hours per experiment. The slide-by-slide version of this report is `presentations/case-ladder/lesto-case-ladder.pdf`.

References

[1] F. Lobresco, G. Tiebel, G. C. Conigliaro, P. Steinegger, O. Marinich, N. I. Prasianakis, J. Neuhausen, B. Ničeno, *Assessment of temperature-based and vapor-pressure-driven models for CFD simulation of volatile species deposition*, Nucl. Eng. Des. 459 (2026) 115201. doi:10.1016/j.nucengdes.2026.115201

[2] L. Liu, V. Zobnin, I. Zivadinovic, R. Eichler, A. Vögele, D. Herrmann, A. Aerts, P. Steinegger, J. Neuhausen, *Chemical speciation of iodine evaporated from liquid lead–bismuth eutectic investigated by thermosublimatography*, J. Radioanal. Nucl. Chem. 334 (2025) 5201–5216. doi:10.1007/s10967-025-10246-4